System context

English version of ../../architecture/system-context.md.

What this is

BrainStorm is a web game/quiz with rooms, a leaderboard, a learning mode, campaigns, user-generated questions and optional AI hints. One Flask + Socket.IO process, data in SQLite inside `DATA_DIR`.

Context diagram

Module boundaries

Layer	Directory	Responsibility
Domain	`app/domain/`	models, registry, game rules — no Flask, no network
HTTP	`app/http_api/`	routes, validation, access
Realtime	`app/realtime/`	rooms, chat, round scheduler
Services	`app/services/`	accounts, achievements, campaigns, leaderboard, UGC
Storage	`app/db.py`, `app/user_db.py`	SQLite, schema, WAL, idempotent migrations
External	`app/ai_client.py`	GigaChat; offline question bank without a key

Rule: the domain imports no Flask, `http_api` holds no game rules.

Failure modes of external systems

Dependency	Mandatory	When missing
Filesystem (`DATA_DIR`)	yes	the application cannot store data
GigaChat	no	the offline question bank is used
eventlet	no	`SOCKETIO_ASYNC_MODE=threading`

Supported scope

Declared: Python 3.12, Linux + Docker/Compose, a browser with WebSocket. Not declared: several instances on a shared SQLite, clustered realtime (rooms live in process memory), other DBMS.